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CTGDiff: A Conditional Diffusion Model for Cardiotocography Signal Synthesis

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***Abstract*—The analysis of Cardiotocography (CTG) signals is often hindered by challenges such as limited data availability and label imbalance, which can undermine the performance of deep learning models. To address these issues, we present CTGDiff, a novel conditional diffusion model designed for generating synthetic Fetal Heart Rate (FHR) and Uterine Contraction (UC) signals. CTGDiff leverages both Phase-Rectified Signal Averaging (PRSA) spectrograms and UC as conditioning inputs for FHR, and integrates time encoding, condition generation from PRSA features, and residual blocks with dilated convolutions to capture both temporal dynamics and long-range dependencies. Extensive experiments, both qualitative and quantitative, demonstrate the model's ability to synthesize high-quality CTG signals. In comparison with GANs and image-based diffusion models, CTGDiff achieves superior signal fidelity and distribution similarity for FHR, as indicated by metrics such as a 0.004 maximum mean deviation (MMD), 0.646 percent root mean square difference (PRD), 3.951 relative entropy (RE), and 0.291 Frechet distance (FD). Expert evaluations confirm that the model can generate both normal and abnormal CTG signals with high accuracy, conditioned on specific input data. These results underscore the potential of diffusion models for a wide range of applications in biomedical time series analysis, including signal synthesis, imputation, and noise reduction.**

***Index Terms*—diffusion models, PRSA, FHR, Cardiotocography**

I. Introduction

Timely monitoring and detection of risks for both the fetuses and pregnant women are essential for preventing major pregnancy complications and are of great importance in public health. Among these techniques, cardiotocography (CTG) has been the standard method for monitoring fetal well-being [1]–[5]. Numerous advanced Computer-Aided Diagnosis (CAD) methods based on AI and deep learning have been developed [6]–[13]. Enhancing the performance of deep learning models in CTG research necessitates a large amount of clinically labeled data. However, obtaining CTG data is challenging due to three main issues: data scarcity, class imbalance, and concerns about patient privacy. As far as we know, there are few available FHR dataset such as CTU-UHB with 552 recordings [14], [15] and CULF-DB with only 155 recordings [16], [17]. Therefore, developing a generative model to synthesize high-quality FHR recordings can effectively address these issues.

Existing deep learning approaches to synthesizing CTG signals are all based on GANs. FHRGAN [18] incorporates an auxiliary classifier to generate multi-category FHR signals from noise, distinguishing between normal and abnormal samples. However, its ability to enhance the diversity of synthetic signals is limited by the use of only two categories and the minor differences in Gaussian noise used as input data. CTG-GAN [19] integrates self-attention and residual modules within a Conditional GAN framework, finely tuned to accurately replicate the complex patterns characteristic of FHR data. Despite effectively capturing overall variability, the model falls short in emulating some highly specific characteristics of CTG signals. There is potential for further enhancement of the model's generalization capabilities, particularly through the utilization of a larger, real-world dataset.

Diffusion models have shown great potential in modeling complex data distributions, particularly in image generation and data synthesis [20], [21]. Inspired by their success in image synthesis, some studies have explored their capability in processing and generating time-series signals. Biodiffusion is a diffusion model tailored for biomedical signal synthesis [22], which was tested on three types of data: simulated signals, accelerometer signals, and ECG signals. Additionally, a conditional diffusion model, TimeDiff [23], was proposed to generate complete multivariate time series from incomplete data. Tosato et al. [24] introduced a class-conditioned diffusion model that generates ECG recordings. Another diffusion model for synthesizing ECG signals was proposed in [25]. However, this model maps 1D ECG data into 2D space and converts them into image files, which results in worse performance compared to the Wassertein GAN with Gradient Penalty (WGAN-GP) [26], which processes input/output data in 1D space.

In this work, we propose a novel diffusion model that uses PRSA curve spectrograms and UC as conditions to generate FHR signals. Our contributions include formulating CTG signal synthesis as a generative denoising process. This is the first

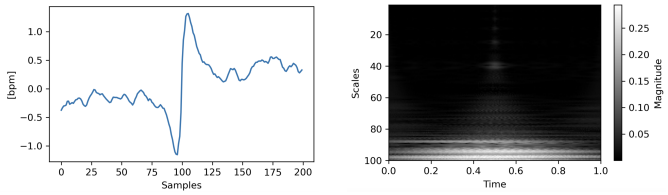
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study to apply the diffusion model to CTG signal synthesis, solving the problem of model collapse and generating high-fidelity data with a distribution similar to real CTG data. We also introduce the PRSA curve for generating paired spectrogram conditions. This method enhances the model's ability to distinguish between normal and pathological features in raw CTG data, providing precise and effective spectral guidance during the signal generation process. Lastly, we conduct a thorough comparison with several state-of-the-art GAN models tailored for CTG signal synthesis. This comparison includes various evaluation metrics to rigorously assess the quality and fidelity of the synthetic data.

II. Method

A. PRSA Curve and Spectrogram

Inspired by [27], we use the Phase-Rectified Signal Averaging (PRSA) spectrogram as the label for each signal. The spectral method derived from PRSA better differentiates high-risk pregnancies compared to other traditional spectrum estimators. We compute the spectrogram from the wavelet coefficients. The PRSA curve and its spectrogram are shown in Fig. 1a and Fig. 1b, respectively.



(a) PRSA curve of FHR series. (b) CWT of PRSA data.

Fig. 1: Comparison of PRSA curve and CWT of PRSA data: (a) PRSA curve and (b) CWT of PRSA data.

In Fig. 1a, the PRSA curve is used to identify deceleration anchor points in the FHR signal, which are critical for detecting and analyzing fetal health conditions. This curve helps in distinguishing between normal and abnormal patterns by averaging the signal around specific anchor points. In Fig. 1b, the CWT is applied to the PRSA curve to convert it into a time-frequency representation, providing a detailed view of the frequency components of the signal over time. This spectrogram aids in identifying specific patterns and anomalies by highlighting the variations in signal frequency that correspond to physiological and pathological changes.

B. Model Architecture

Our model is designed to jointly train and generate multivariate signals of FHR and UC within a mixed training framework. The network structure is illustrated in Fig. 2, and the training process involves using both raw signals and their transformed spectral representations as conditions. For UC, the raw UC signal is processed by adding noise, while its spectrogram representation is obtained through Continuous Wavelet Transform (CWT) and used as a conditioning input. For FHR, the raw signal is also perturbed with noise, and its transformation into a Phase-Rectified Signal Averaging (PRSA) curve, followed by a CWT transformation into a spectrogram, serves as **Condition 1**. Additionally, the corresponding UC signal is incorporated as **Condition 2** to better capture dependencies between FHR and UC.

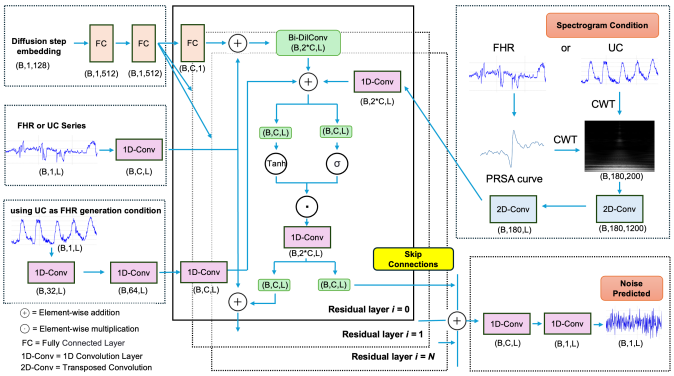


Fig. 2: Network Architecture of CTGDiff for Signal Synthesis.

During the generation phase, UC signals are synthesized by using randomly sampled noise as input, with the UC spectrogram acting as a condition to guide the process. For FHR generation, the previously synthetic UC signal serves as the first condition, while the FHR spectrogram (transformed from PRSA and CWT) is used as the second condition. These conditions provide structural guidance, ensuring the generated FHR signal maintains consistency with physiological patterns. The model employs residual connections, dilated convolutions, and conditioning mechanisms to effectively capture both temporal and spectral dependencies, enabling accurate signal generation.

As described in Section II-A, each FHR series has a corresponding PRSA curve spectrogram that serves as guidance for generation in this model. After pre-processing to extract the PRSA curve and applying the CWT to create the spectrogram, the spectrogram is upsampled through two layers of transposed 2-D convolutions. Subsequently, a 1×1 convolutional layer is applied to map the dimension to 2 residual channels.

III. Experiment

A. Dataset

This study utilized the CTU-CHB Intrapartum Cardiotocography dataset. The dataset comprises a collection of 552 CTGs from the University Hospital in Brno (UHB) and the Czech Technical University (CTU). Each recording lasts up to 90 minutes and includes a FHR time series, a uterine contraction signal, and detailed maternal, delivery, and fetal clinical information. The signals are sampled at 4 Hz.

B. Pre-processing

A series of signal processing algorithms were applied to the raw signals to eliminate the effects of missing values, baseline wandering, and artifacts. Initially, gaps longer than 60 samples (15 seconds) in the FHR time series were removed. Linear

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interpolation was then employed for the FHR signal, and signals with more than 10% missing values were excluded from the model inputs. A detrend function was used to eliminate baseline drift. Finally, the long signals were segmented into 20-minute intervals, each containing 4,800 data points. Then the signals were downsampled to 2,400 points.

C. Evaluation Results

In line with previous work, we measured and compared the fidelity of the synthetic data using various evaluation metrics. To assess the effectiveness of the PRSA curve spectrogram in guiding the model to generate normal and abnormal FHR signals, an ablation study including visual inspection and Principal Component Analysis (PCA) was conducted.

1) Fidelity: High fidelity indicates that the synthetic signals have a distribution similar to the real CTG signals. Several evaluation metrics were applied to determine whether the synthetic data distribution $z \sim P_z$ is indistinguishable from the real data distribution $x \sim P_{\text{data}}$. Relative Entropy (RE), also known as Kullback-Leibler (KL) Divergence, and Frechet Distance (FD) provide different perspectives on whether the synthetic data capture the characteristics of the real data.

FD measures the similarity between two distributions by finding the minimum possible distance between the two continuous signals. Lower RE and FD values indicate better quality of the synthetic signal. Maximum Mean Discrepancy (MMD), Sliced Wasserstein Distance (SWD), and Percent Root Mean Square Difference (PRD) are also statistical tests used to measure the difference between two different distributions. SWD is computed by averaging the Wasserstein distance over multiple 1-dimensional projections. PRD quantifies the relative difference between two sets of signals.

The evaluation results are shown in Table I. We re-trained six GAN-based architectures to generate FHR signals as baselines: SeqGAN [28], DCGAN [29], CGAN [30], CWGAN [31], FHRGAN [18], and CTGGAN [19]. Furthermore, we optimized a 2D image-based diffusion model. The one-dimensional (1D) FHR signal was embedded into a two-dimensional (2D) space by mapping the time series signal from Cartesian coordinates to polar coordinates. Gramian Angular Summation/Difference Field (GASF/GADF) and Markov Transition Fields (MTF) were used to construct three separate 2D matrix embeddings of the time series [32]. We then designed a diffusion model with Convolutional Neural Networks (CNNs) as the backbone, incorporating residual connections and an attention mechanism. The details of the original model are referred in [25].

Based on the evaluation metrics presented in Table I, our method CTGDiff demonstrates superior performance compared to existing models. CTGDiff achieves the lowest RE of 3.951 and the best FD of 0.291, outperforming all GAN models. In terms of MMD, CTGDiff reaches an impressive 0.004, indicating an exceptional match between the synthetic and real FHR data distributions. Furthermore, the results for CTGDiff, specifically in terms of UC, reveal a remarkable performance with an MMD of 0.004, a low RE of 0.939, and a PRD of 0.776, showing that CTGDiff effectively captures the distributional characteristics and fine-grained features of UC signals. Although the SWD is marginally lower for CTGGAN (0.012), CTGDiff still maintains competitive performance with an SWD of 0.245. Moreover, CTGDiff attains the best PRD score of 0.646, further demonstrating its capability to generate high-fidelity synthetic data with minimal deviation from the true distribution. Overall, these results underscore CTGDiff's effectiveness in capturing both the overall structure and fine-grained distributional characteristics of the CTG data.

TABLE I: Comparison of Different Models for Synthesis

	Evaluation Metrics				
Model	RE(↓)	FD(↓)	MMD(↓)	SWD(↓)	PRD(↓)
SeqGAN [28]	10.324	0.974	6.527	0.541	10.635
DCGAN [29]	9.532	0.795	5.267	0.254	10.431
CGAN [30]	9.012	0.763	5.022	0.254	9.431
CWGAN [31]	8.523	0.715	4.568	0.173	0.702
FHRGAN [18]	8.059	0.682	1.975	0.086	5.437
CTGGAN [19]	5.621	0.614	0.215	**0.012**	4.821
DDPM [25]	7.059	0.682	6.323	0.521	9.786
CTGDiff	**3.951**	**0.291**	**0.004**	0.245	**0.646**

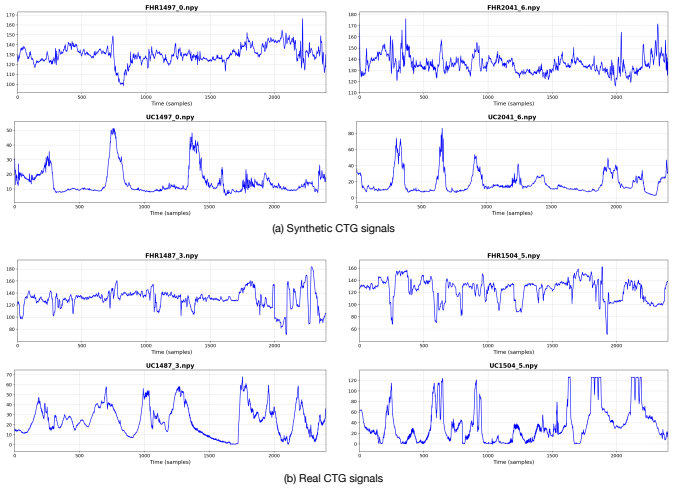


Fig. 3: Comparison of Synthetic Signal and Real CTG Signal.

These results highlight the ability of CTGDiff to generate high-fidelity synthetic FHR signals that closely align with real-world data. Fig. 3 further illustrates this comparison, where (a) represents the synthetic signals, and (b) corresponds to real CTG signals, showcasing their structural similarities and differences.

2) Classification Accuracy: We evaluate the ability of our model to generate synthetic data that closely matches real data, focusing on its capability to generate different FHR signals classified as *normal* or *abnormal*. Given the variability in signal quality in the CTU-CHB dataset—where some signals are recorded during labor and others via ultrasound instead of direct scalp electrode—the signal quality is inconsistent, with some containing significant noise. These factors complicate the use of machine learning for classification tasks.

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To address this, we included evaluations of 160 synthetic samples by three obstetricians in our study. We extracted PRSA curve spectrograms from the test set, which served as conditions for our trained model to generate diverse data. We then calculated the accuracy, precision, specificity, sensitivity, and negative predictive value of the classification.

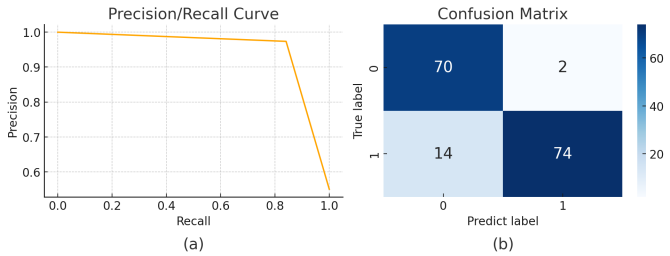


Fig. 4: (a) Precision/Recall Curve and (b) Confusion Matrix for Manually Classification by An Obstetrician.

Fig. 4 illustrates the performance of the classification model with (a) the Precision/Recall curve, and (b) the Confusion Matrix. The marked values demonstrate that our model can accurately generate different classes of data based on the PRSA curve spectrogram condition.

TABLE II: Classification Accuracy with Different Ratios of Real and Synthetic CTG Signals

No.	Real Signal	Synthetic Signal	Accuracy(↑)
1	100%	0%	75.0%
2	90%	10%	73.0%
3	80%	20%	73.0%
4	70%	30%	69.0%
5	60%	40%	67.1%
6	50%	50%	68.5%

Table II presents the classification accuracies achieved by progressively substituting real CTG signals with synthetic counterparts during the training of a attention-based CNN-BiLSTM classification model [33]. The model's performance was assessed using a test set composed exclusively of original signals. The ratios of real to synthetic signals varied from 100% real (and 0% synthetic) to 50% real (and 50% synthetic).

D. Evaluation of Conditional and Unconditional Models

Our ablation experiments were designed to explore the hypothesis that conditional diffusion models, when provided with PRSA spectrogram data, produce synthetic FHR signals with greater diversity and closer alignment to physiological norms compared to unconditional models.

We conducted a visual inspection to assess the diversity of FHR signal samples generated by unconditional and conditional DDPM models. As depicted in Fig. 5, the incorporation of PRSA spectrogram conditions in the DDPM model notably enriches the variability of the synthetic data. It is evident that the data generated by unconditional DDPM exhibit similar patterns at the beginning and end of the signals, while the conditional approach introduces a broader range of patterns.

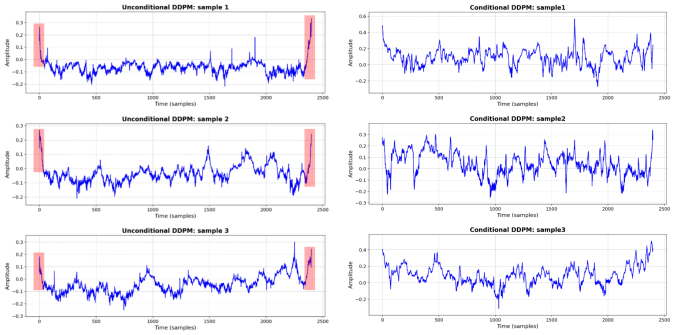


Fig. 5: Visual Comparison of FHR Signal Samples Generated by Unconditional and Conditional DDPM Models.

These findings highlight the importance of incorporating PRSA spectrogram conditions in the diffusion process, demonstrating that conditional models generate more diverse and physiologically realistic FHR signals. By leveraging conditional information, the model effectively captures a wider range of FHR signal variations, which is crucial for generating high-fidelity synthetic data.

IV. Discussion and Limitation

Despite the promising results, our study lacks a systematic analysis of the correlation between FHR and UC. One potential approach to address this limitation is to conduct a causal analysis between FHR and UC, which would provide deeper insights into their interdependencies. Another approach involves training a classification model where the input consists of FHR and UC from the same real CTG signal, labeled as 1, and FHR and UC from different CTG signals, labeled as 0. This model can then be used to infer whether the FHR and UC in synthetic CTG signals are consistent by verifying whether the predicted label is 1 or 0. Both of these methodologies could be explored to enhance the evaluation of synthetic CTG signals and further validate their physiological consistency.

V. Conclusion

This study introduced a novel diffusion probabilistic model conditioned on PRSA spectrograms for synthesizing CTG signals. Our model captures essential physiological and pathological characteristics, generating high-fidelity and diverse data. Utilizing the CTU-CHB dataset, we demonstrated superior performance over GAN-based models and a 2D diffusion model across multiple metrics. Expert validation confirms the clinical realism of our synthetic signals. The model's success in mimicking FHR signals suggests significant potential for improving fetal health monitoring and diagnostic tools.

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